

REDEMPTION CITY NAVIGATOR

Architecture & System Design Document

Offline-first navigation, market access and local service delivery for Redemption City.

Executive Summary

Redemption City Navigator is a mobile-first platform built to make movement, discovery and local access inside Redemption City easier. It brings together map-based navigation, offline place search, chalet discovery, vendor access, delivery support, Sunday Shuttle and local service information in one practical app experience.

The product is designed for people who need to move through Redemption City with confidence: first-time visitors, worshippers, residents, vendors, chalet guests, students, event attendees and families. Instead of depending on repeated verbal directions or scattered information, users can search, locate, order, move and connect through one organized mobile interface.

Product Positioning

Redemption City Navigator is a practical city-scale companion for Redemption City: navigation, local commerce, mobility and doorstep delivery brought into one mobile experience.

PRODUCT	Redemption City Navigator
CATEGORY	Safety, navigation, local services and community commerce
PLATFORM	Expo React Native mobile app with TypeScript
CURRENT STATE	Working mobile MVP with local search, map display, route data, market, chalets, shuttle, vendor and delivery-related screens.
PRIMARY USERS	Visitors, worshippers, residents, vendors, chalet guests, delivery riders and camp service operators.
REPOSITORY	https://github.com/Supreme5050/redemption-city-navigator

1. Problem and Product Value

Redemption City is large, active and constantly moving. People come in for services, conventions, accommodation, meetings, shopping and family visits. Many of them know where they want to go, but do not know the best road, nearest landmark, available vendor or most convenient way to get there.

The result is familiar: people ask several strangers for direction, walk longer than necessary, miss pickup points, struggle to locate chalets, or find it difficult to buy items from nearby vendors. Redemption City Navigator turns that scattered local knowledge into a clear mobile system.

- Visitors find roads, landmarks and service points faster.
- Residents and worshippers get a dependable local discovery tool.
- Vendors gain visibility through Market Hub and delivery support.
- Chalet users can access accommodation information from a dedicated section.
- Delivery agents and local helpers gain a structured way to earn within Redemption City.
- Operators and partners get a scalable base for verified local information.

Why the Solution Matters

The app addresses a real and repeatable problem inside a real location. It is not a generic map idea; it is built around Redemption City movement, services, commerce and local logistics.

2. Product Readiness Snapshot

- The mobile app runs in Expo and is available as a working prototype for demonstration.
- Core tabs and screens are implemented with Expo Router.
- Local place search is connected to structured Redemption City data.
- The map screen supports hybrid/satellite view, destination markers and saved route corridors.
- Route data is separated from UI code, making coordinate updates safe and controlled.
- Market Hub, vendor screens, delivery tracking, proof flows, chalets and shuttle sections are present in the source code.
- The public GitHub repository provides full review access to the codebase.

3. Core User Experience

1. The user opens the mobile app and lands inside a simple tab-based experience.
2. The user searches for a road, chalet, church, vendor, product or service point.
3. The app resolves the search from its local Redemption City dataset.
4. The map displays the destination, route corridor and current-location marker in a clear visual view.
5. The user can move to Market Hub to find vendors and products inside Redemption City.
6. When an item is ordered from a vendor, the delivery flow supports movement from vendor to customer doorstep.
7. The user can also access Chalets and Sunday Shuttle sections without leaving the app.

4. System Architecture

The system is built with a clean separation between the mobile interface, local data, map rendering, route data and service modules. This makes the product easy to review and easier to improve without breaking working screens.

Layer	Role in the System	Implementation
Mobile Interface	Presents the app screens, tabs, map, search, marketplace, chalet and shuttle flows.	Expo React Native, Expo Router, reusable typed components.
Location & Maps	Shows destination markers, current position and saved internal road corridors in satellite or hybrid 3D view.	react-native-maps, Google provider, Expo Location, local route coordinates.
Local Data Layer	Keeps important Redemption City roads, places, vendors, chalets and service entries available in the app.	Typed TypeScript datasets: places, routes, vendors, products and chalets.
Route Engine	Selects the best saved corridor for a destination and draws it as a polyline.	verifiedRoutes.ts plus route-selection and coordinate-calibration workflow.
Commerce Layer	Connects users to vendors, products, protected orders and delivery flows.	Market, vendor, products, orders, delivery tracking and proof screens.
Evidence Layer	Makes the work reviewable through source code, mobile demo, screenshots and documentation.	Public GitHub repository, demo video, PDFs and screenshots.

5. Capability Modules

Module	What Users Can Do	Technical Basis	Review Evidence
Map & Navigation	Search locations, view destination markers and follow saved internal route corridors.	places.ts, verifiedRoutes.ts, map screen logic.	3D/hybrid map screenshots and source code.
Offline Place Search	Find roads, buildings, churches, offices and known landmarks even with weak network.	Typed local place dataset with aliases and address hints.	Search works from structured local data.
Chalets	Access a dedicated accommodation discovery section for Redemption City hospitality use.	Chalet screens and data structure.	Chalets tab and chalet detail flow.
Market Hub	Discover vendors, products and local business activity inside Redemption City.	Vendor, products and market screens.	Market tab and vendor screens in repository.
Delivery System	Move customer orders from vendor location to the customer doorstep, with tracking/proof flow.	Delivery tracking, proof, protected order and order status screens.	Delivery files exist in the mobile app source.
Sunday Shuttle	Support Sunday mobility by organizing ride/shuttle discovery around the camp movement pattern.	Sunday Shuttle screen and route-ready module.	Dedicated Sunday Shuttle source file.
Vendor Center	Give sellers a structured area for orders, products, wallet and customer interaction.	Vendor dashboard, orders, products, wallet and chat screens.	Vendor module files in app source.
Feedback & Trust	Support user confidence through feedback, dispute, proof and protected order flows.	Feedback reports, dispute and delivery proof screens.	Trust-related files exist in codebase.

6. Delivery and Job Creation Model

The delivery section is one of the strongest parts of the service model. When a user orders an item from a vendor in Market Hub, the platform supports a flow where the vendor prepares the order, a delivery person picks it up, and the item is delivered to the customer doorstep inside Redemption City.

This does more than make shopping convenient. It creates a local work layer. Students, residents and approved delivery helpers can earn by completing deliveries within the camp. Vendors get more sales, customers receive items without unnecessary movement, and the app becomes part of the local economy rather than just a direction tool.

CUSTOMER VALUE	Users can order from local vendors and receive items at their doorstep.
VENDOR VALUE	Vendors gain visibility, orders and a delivery channel beyond walk-in customers.
DELIVERY AGENT VALUE	Local riders/helpers can earn from verified delivery tasks inside Redemption City.
TRUST LAYER	Delivery tracking, proof of delivery and protected order screens support safer transactions.
COMMUNITY IMPACT	The platform supports convenience, local trade and job creation inside Redemption City.

7. Offline-First Design

The app treats local information as a product asset. Important roads, places, route corridors, vendors and services are stored as structured data in the codebase. This gives the app a reliable base even when public map data is incomplete or internet access is unstable.

- Local datasets reduce dependence on live internet search.
- Aliases make search easier when users type shortened or familiar names.
- Route coordinates are managed separately from the map interface.
- Satellite/hybrid map view helps verify road and building context.
- The structure is ready for database synchronization while keeping offline resilience.

Coordinate Accuracy Approach

Routes are treated as saved corridors made up of road and junction points. As coordinates are verified, the route file can be updated without redesigning the app.

8. Navigation and Route Model

The navigation model is designed for a place like Redemption City, where public map data may not always capture internal roads correctly. The app uses saved route corridors and displays them directly on the map. This gives the team control over route quality and makes continuous improvement possible.

DESTINATION DATA	places.ts stores location names, coordinates, categories, zones and aliases.
ROUTE DATA	verifiedRoutes.ts stores named route corridors and coordinate sequences.
RENDERING	react-native-maps draws route polylines, markers and satellite/hybrid map context.
CALIBRATION	Road bends and junctions can be collected from satellite view and added to route data.
REVIEW VALUE	Judges can see both the route behavior and the code structure behind it.

9. Technical Stack

MOBILE FRAMEWORK	Expo React Native
LANGUAGE	TypeScript
NAVIGATION	Expo Router
MAP LAYER	react-native-maps with Google provider
LOCATION	expo-location
DATA MODEL	Typed local datasets for places, routes, vendors, products and chalets
VERSION CONTROL	Git and GitHub
PROTOTYPE EVIDENCE	Expo mobile demo, screenshots, public repository and PDF documentation

10. Security, Trust and Operational Readiness

- The MVP does not expose sensitive backend credentials in the client code.
- Local datasets make the app reviewable and stable during demonstration.
- Protected order, delivery proof and dispute screens provide a base for transaction trust.
- The codebase is modular, so backend services can be connected without rebuilding the whole app.
- The map route data can be improved gradually through verified road coordinates.

Operational Readiness

The current app is strong enough to demonstrate the product direction, user experience and technical foundation. Its structure supports real deployment work, vendor onboarding and route-data improvement.

11. Review Evidence

The submission is supported by a public repository, mobile demo evidence, screenshots and product documentation. Together, these materials show that the solution has moved beyond an idea into a working mobile MVP.

GITHUB REPOSITORY	https://github.com/Supreme5050/redemption-city-navigator
PROTOTYPE EVIDENCE	Mobile demo video / Expo demonstration link supplied in the submission portal.
SCREENSHOTS	Mobile screenshots supplied in the submission portal.
ARCHITECTURE DOCUMENT	This document.
BUSINESS / GTM DOCUMENT	Submitted as a separate PDF.
PITCH DECK	Submitted as a separate PDF.

12. Impact and Scale

Redemption City Navigator is valuable because it serves an environment where movement, accommodation, commerce and local knowledge are already daily needs. The app organizes those needs into one mobile product and makes them easier to access.

- Visitors reach locations faster and with less confusion.
- Worshippers and residents get one reliable tool for local movement.
- Vendors are easier to discover and can receive orders beyond walk-in traffic.
- Delivery agents can earn through structured doorstep delivery tasks.
- Chalet and hospitality discovery becomes easier for guests.
- The same model can serve other large camps, universities, estates, retreat grounds and event cities.

Closing Statement

Redemption City Navigator combines a clear local problem, a working mobile MVP, a public codebase and a practical service model. It gives Redemption City a foundation for smarter navigation, stronger local commerce and organized doorstep delivery.